

# MongoDB Exercise

## 1. \*\*Create Operation:\*\*

- Create a new database named "library" and switch to it.

```
Microsoft Windows [Version 10.0.26100.8246]
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C:\Users\harsh>mongosh
Current Mongosh Log ID: 69f71b3c45ccf3285babc113
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=3000&appName=mongosh+2.8.3
Using MongoDB:  5.2.7
Using Mongosh:  2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-05-02T08:30:04.804+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use library
switched to db library
```

- Create a collection named "books" with fields: title, author, and published\_year.

```
library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertOne({title: "Harry Potter", author: "J.K. Rowling", published_year: 1997})
{
  acknowledged: true,
  insertedId: ObjectId('69f71b6145ccf3285babc114')
}
```

- Insert a document into the "books" collection with the details of your favorite book.

```
library> db.books.insertMany([
  { title: "1984", author: "George Orwell", published_year: 1949 },
  { title: "The Catcher in the Rye", author: "J.D. Salinger", published_year: 1951 },
  { title: "MongoDB Basics", author: "John Doe", published_year: 2022 },
  { title: "NoSQL Guide", author: "Jane Smith", published_year: 2021 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f71b8c45ccf3285babc115'),
    '1': ObjectId('69f71b8c45ccf3285babc116'),
    '2': ObjectId('69f71b8c45ccf3285babc117'),
    '3': ObjectId('69f71b8c45ccf3285babc118')
  }
}
```

## 2. \*\*Read Operation:\*\*

- Retrieve all documents from the "books" collection.

```
library> db.books.find()
[
  {
    _id: ObjectId('69f71b6145ccf3285babc114'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc115'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021
  }
]
```

- Find and display only the documents where the author is "J.K. Rowling".

```
library> db.books.find({ author: "J.K. Rowling" })
[
  {
    _id: ObjectId('69f71b6145ccf3285babc114'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    published_year: 1997
  }
]
```

- Fetch and display the document with the earliest published year.

```
library> db.books.find().sort({ published_year: 1 }).limit(1)
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc115'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  }
]
```

### 3. \*\*Update Operation:\*\*

- Update the published year of the book with the title "The Catcher in the Rye" to the current year.

```
library> db.books.updateOne({ title: "The Catcher in the Rye" }, { $set: { published_year: 2026 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- Add a new field "genre" with the value "Mystery" to all documents in the "books" collection.

```
library> db.books.updateMany({}, { $set: { genre: "Mystery" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

#### 4. \*\*Delete Operation:\*\*

- Remove the document with the title "1984" from the "books" collection.

```
library> db.books.deleteOne({ title: "1984" })
{ acknowledged: true, deletedCount: 1 }
library> db.books.find()
[
  {
    _id: ObjectId('69f71b6145ccf3285babc114'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    published_year: 1997,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
```

- Delete all documents from the "books" collection where the published year is before 2000.

```
library> db.books.deleteMany({ published_year: { $lt: 2000 } })
{ acknowledged: true, deletedCount: 1 }
library> db.books.find()
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
```

## 5. \*\*Advanced Query:\*\*

- Find and display the top 3 recently published books from the "books" collection.

Type 1

```
library> db.books.find().sort({ published_year: -1 }).limit(3)
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
```

Type 2

```
library> db.books.aggregate([{$sort: { published_year: -1 }}, {$limit: 3 }])
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> |
```

### Type 3

```
library> db.books.find({}, { title: 1, published_year: 1, _id: 0 }).sort({ published_year: -1 }).limit(3)
[
  { title: 'The Catcher in the Rye', published_year: 2026 },
  { title: 'MongoDB Basics', published_year: 2022 },
  { title: 'NoSQL Guide', published_year: 2021 }
]
library> |
```

- Retrieve documents from the "books" collection where the title contains the word "MongoDB" or "NoSQL".

### Type 1

```
library> db.books.find({$or: [{ title: /MongoDB/i }, { title: /NoSQL/i }]}))
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
```

### Type 2

```
library> db.books.find({title: { $in: [/MongoDB/i, /NoSQL/i] }})
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
```

### Type 3

```
library> db.books.aggregate([{$match: {title: { $regex: "MongoDB|NoSQL", $options: "i" }}}])
[
  {
    _id: ObjectId('69f71b8c45ccf3285babc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f71b8c45ccf3285babc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> |
```